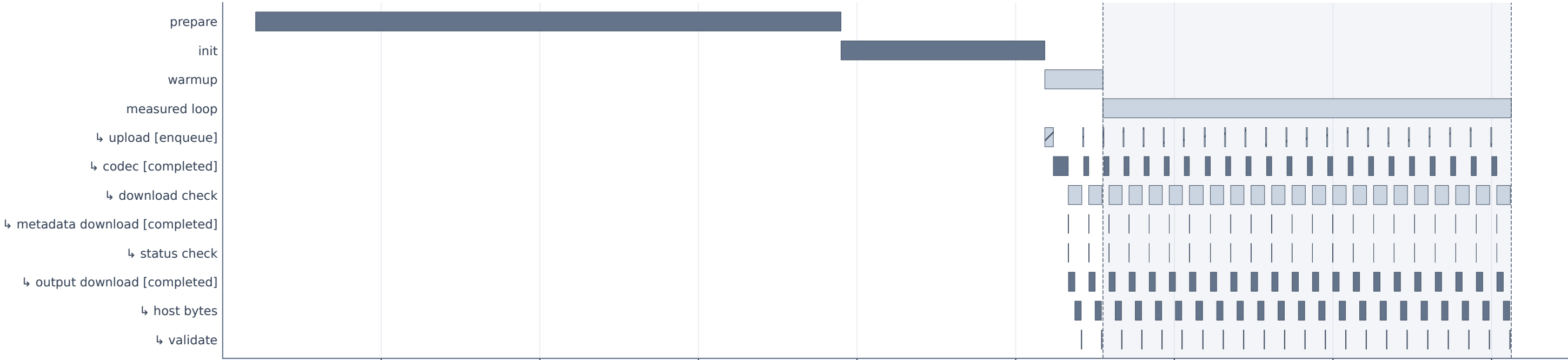


# Decompression workflow · whole process

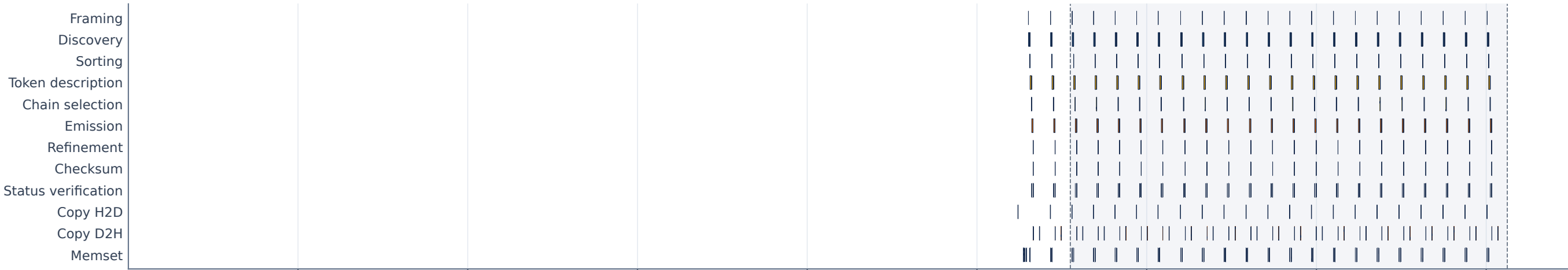
NVIDIA GeForce RTX 3090 · 64 MiB float32 payload · 20 validated measured iterations · source d323fee21565

Instrumented diagnostics · 2 warmups · exact CUDA intervals and sampled process resources share elapsed time

## Host annotations · containers and nested download/check phases



## Exact recorded CUDA intervals · decompression stages



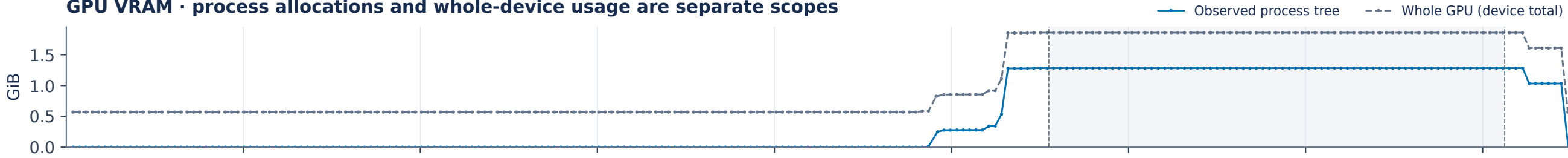
## Copy bytes assigned at completion · stacked by transfer direction



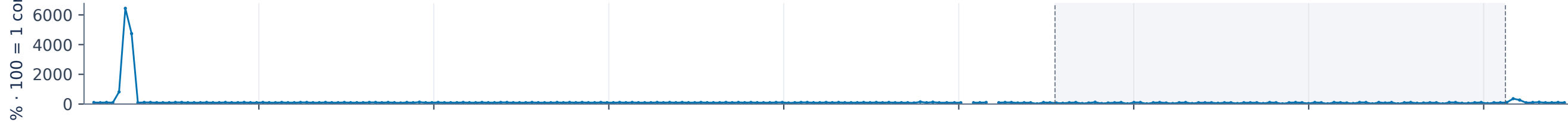
## Recorded kernel activity fraction · interval union / bin duration; not occupancy



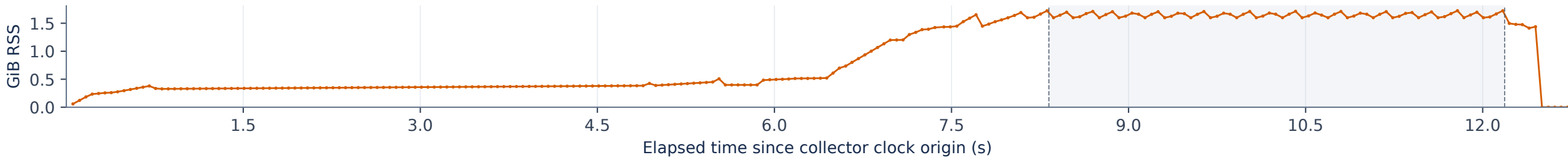
## GPU VRAM · process allocations and whole-device usage are separate scopes



## Process-tree CPU · cumulative CPU-time differences at actual sample intervals



## Process-tree resident memory · summed RSS can count shared pages more than once



Dashed lines bound the measured loop. CUDA bins: 10 ms; view-boundary bins use their actual clipped width. Copies show completed bytes, not bandwidth.

Collector cadence: median 53.876 ms (53.352–65.386 ms). Metric points use query midpoints; CPU uses averaging-interval midpoints.

Host/CUDA alignment:  $\pm 58.907 \mu\text{s}$  (half the origin-mark bracket). Host ranges can nest; they do not establish exclusive GPU ownership or costs.

Download, status validation, host byte construction and output validation are distinct host phases. Coarse resources cannot resolve individual short phases; nulls remain gaps.

Nsight warns potentially missing NVTX events; all expected host phase ranges were validated. Other warnings remain in the report and manifest.